SkySentry AI

From False Alerts to Intelligent Aerial Defense

Visual Verification Layer for Intelligent Aerial Surveillance

Presented by :

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The Problem

High False Positives

Radar systems frequently misclassify birds, debris, and environmental noise as threats, generating excessive false alerts and reducing system reliability.

Low Detectability of Small Drones

Small UAVs have extremely low Radar Cross Section (RCS), making them difficult to detect and distinguish from other small aerial objects.

Operator Alert Fatigue

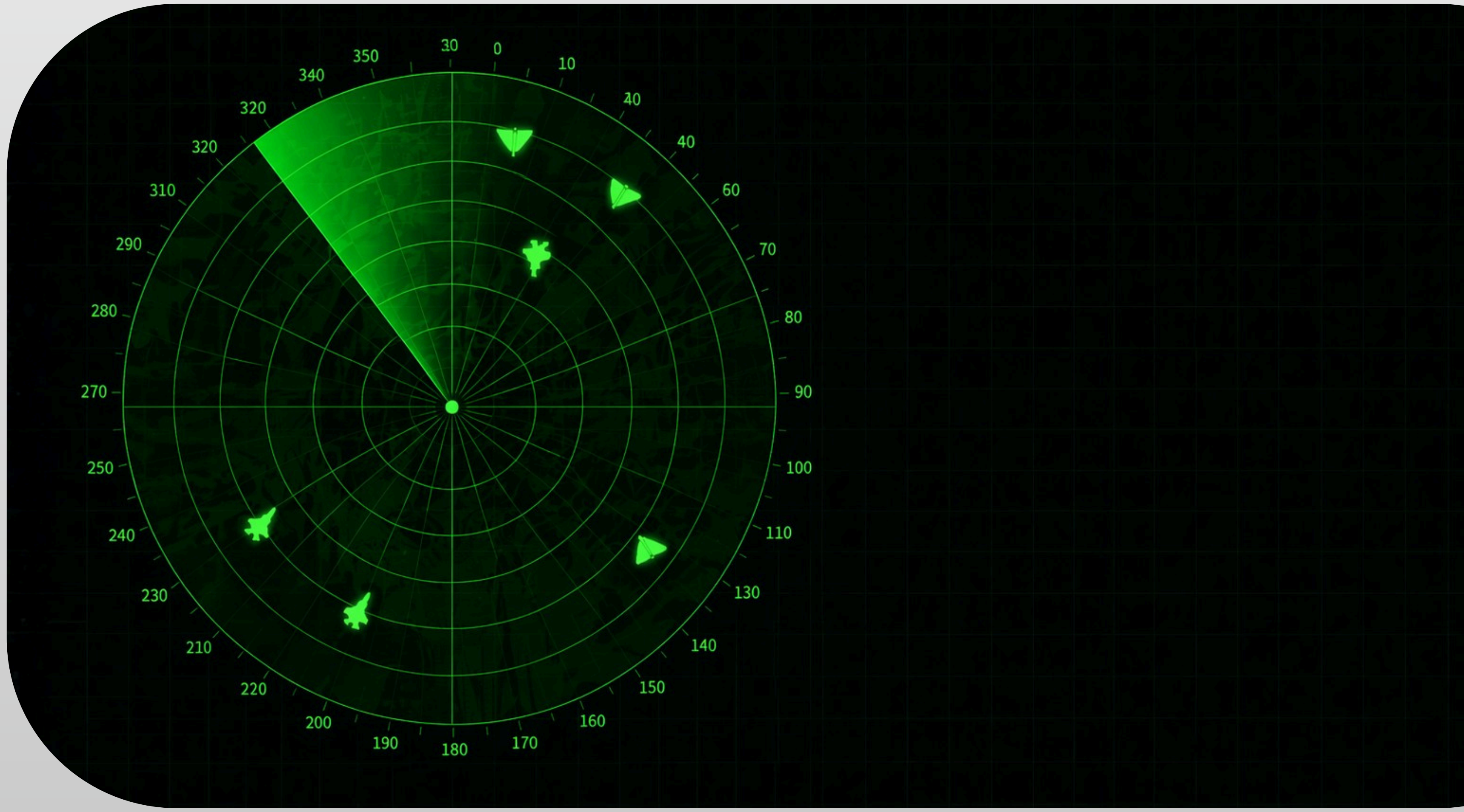
Continuous false alarms overwhelm operators, leading to reduced attention, delayed response times, and increased risk of missing real threats.

Lack of Visual Confirmation

Radar-only detection lacks real-time visual verification, creating uncertainty and slowing down critical decision-making.

Real-World Consequences of Misidentification

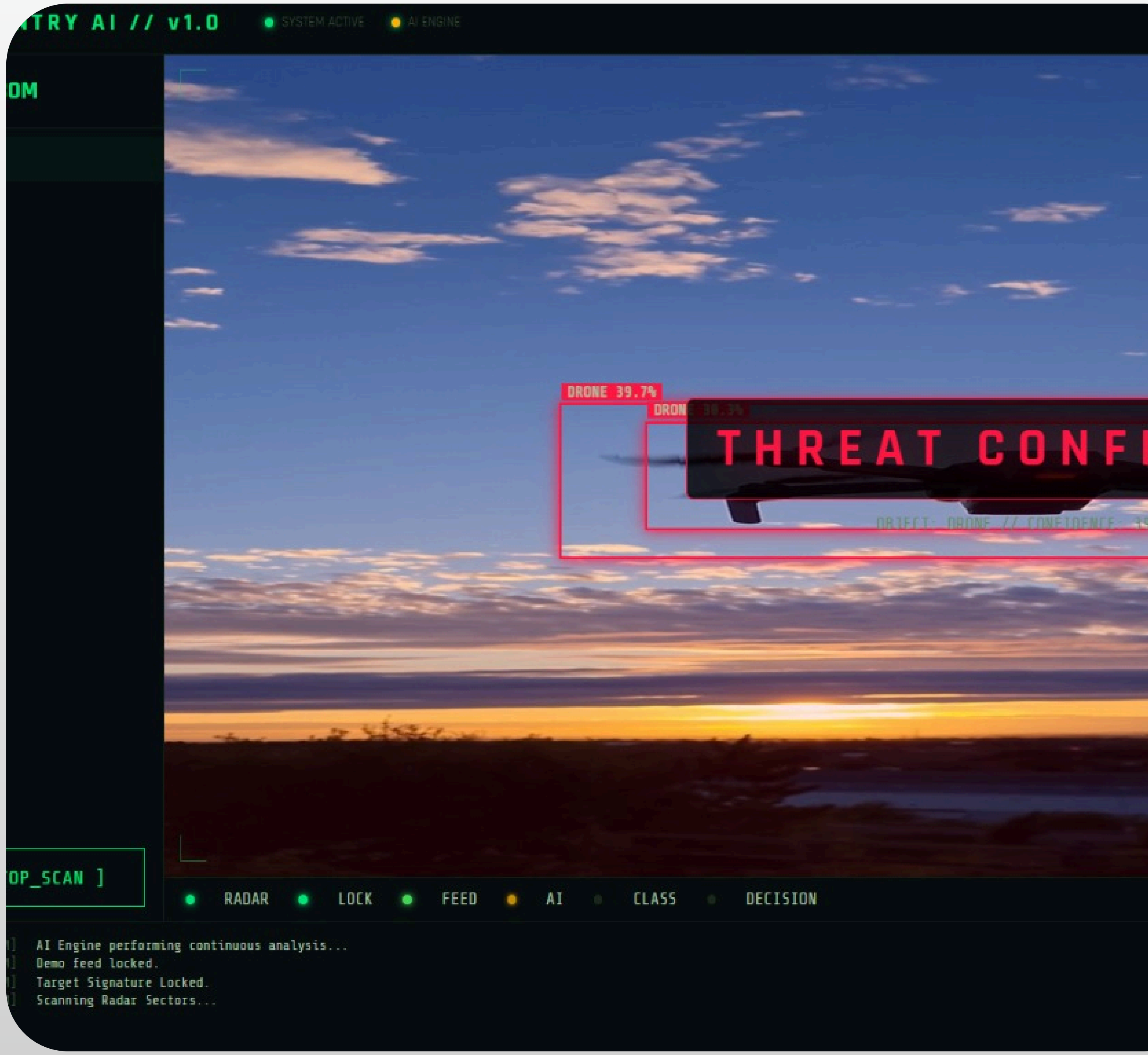
- **IAF Mi-17 Incident (2019)** — Friendly fire due to lack of verification
- **Ukraine Flight PS752 (2020)** — Civilian aircraft misidentified as a threat
- **Military Friendly Fire Cases** — Radar ambiguity leading to incorrect targeting



Existing systems detect objects — but fail to verify whether they are real threats.

Our Solution

SkySentry AI — *Visual Verification Layer for Aerial Surveillance*



- Integrates with radar & camera systems via slew-to-cue mechanism
- Uses **YOLOv10** to detect & classify aerial objects in **real time**
- Filters non-threats (birds, noise) and identifies drones & aircraft
- Converts radar signals into visually verified, **actionable intelligence**
- Sends instant alerts to command centers via REST APIs

SkySentry

Works alongside existing radar systems — no infrastructure replacement required.

how it works?

SkySentry AI — *Visual Verification Layer for Aerial Surveillance*

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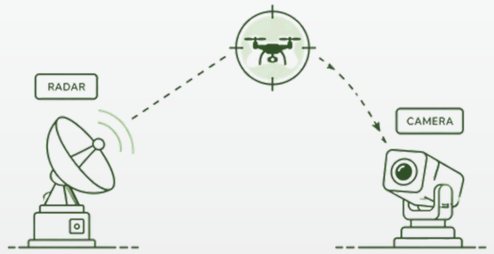
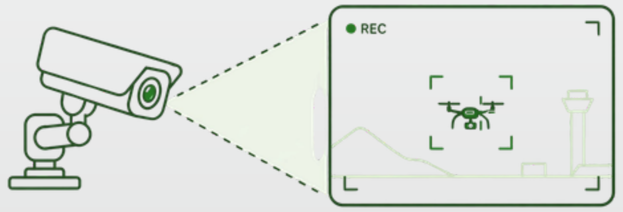


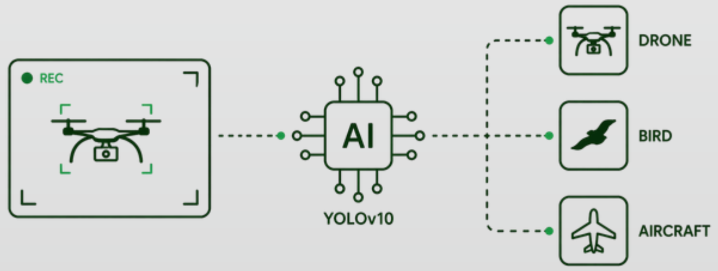
Detect

Radar identifies object and slews camera to target

Capture

Live video feed captured and preprocessed using OpenCV



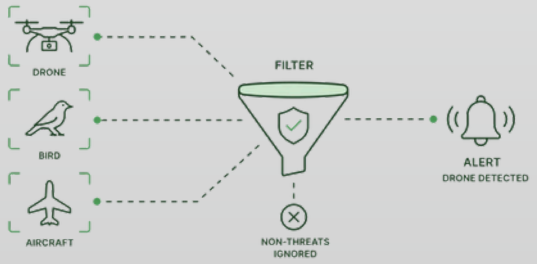


Classify

YOLOv10 detects and classifies: Drone, Bird, Aircraft

Decide

System filters non-threats and triggers alerts for drones



Alert

Dashboard updates + API push + automatic logging

From detection to decision in milliseconds.

Key Features & APPLICATIONS

- **82% Precision**
  Accurate classification of drones, aircraft, and birds
- **Ultra-Fast Inference (<20ms)**
  Real-time processing on RTX 4060
- **RESTful API Integration**
  Instant edge → command center communication
- **Automated Logging**
  Chronological records for post-incident analysis
- **Slew-to-Cue Ready**
  Seamless integration with radar & acoustic sensors

- **Border Surveillance**
  Detect unauthorized drone intrusions
- **Airport Safety**
  Differentiate UAVs from civil aircraft
- **Critical Infrastructure**
  Protect power grids, refineries & govt facilities
- **Military Forward Posts**
  Affordable edge-AI deployment
- **Disaster Response**
  Track drones & unknown aerial objects in rescue ops

SkySentry

Tech Stack

AI / ML

YOLOv10

OpenCV

Backend

Frontend

Dev Tools

CVAT

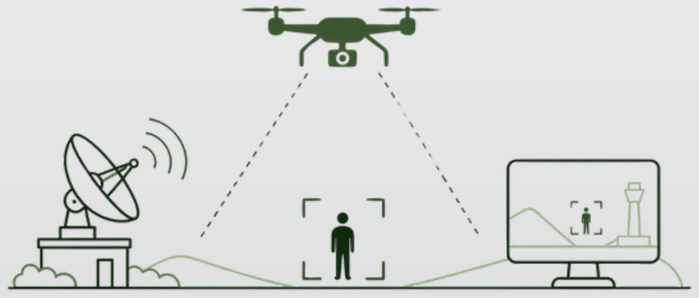
A robust stack powering real-time intelligence.

What Makes SkySentry Different

Turning detection into verified, actionable intelligence.

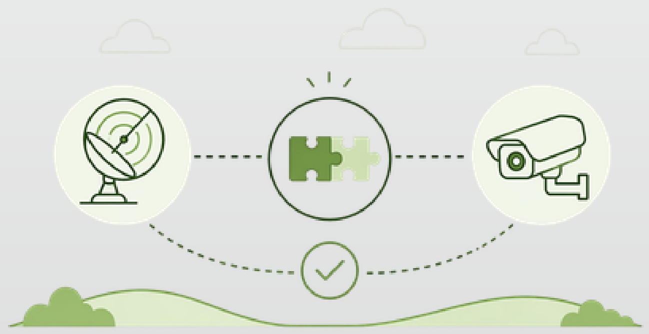
Visual Verification Layer

First system to combine AI vision with radar, converting uncertain detections into confirmed threats



Seamless Integration

Works with existing radar & camera systems — no infrastructure replacement

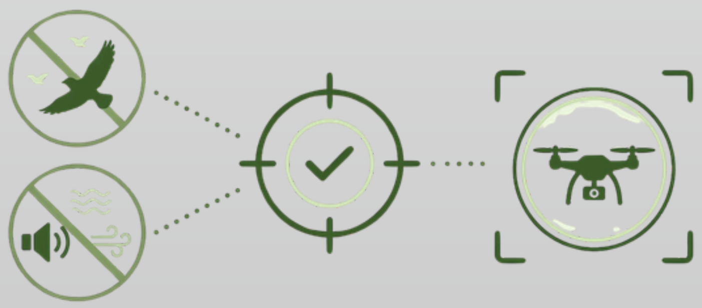


Consumer Hardware, Military-Grade Results

Sub-20ms inference on RTX 4060 — no expensive defense hardware required

Eliminates Alert Fatigue

82% precision significantly reduces false positives from birds & environmental noise



Accessible & Scalable Defense

Delivers high-fidelity aerial monitoring for borders, airports & critical infrastructure

Bridging the gap between detection and decision.

CONCLUSION

SkySentry AI : Detection → Verification → Action

Bridging the gap between raw detection and real-time decision-making.

- Reduces false alerts through AI-driven visual verification
- Enhances decision accuracy and response speed
- Delivers real-time performance on affordable hardware
- Scalable and ready for real-world deployment

SkySentry

OUR TEAM

SkySentry AI — *Visual Verification Layer for Aerial Surveillance*

Pranjal Ranjan — Frontend Architecture
Savyam Shukla — Backend Architecture
Aryan Chauhan — System Research
Naitik Nigotiya — Experience Design
Vaibhavi Utage — Data Pipeline